

Technology #74

Coat the external ventricular drain catheter with a local, sustained■release albendazole formulation to reduce proteinaceous debris and clot formation within the catheter lumen

Engineering Technology-Transfer Dossier

CONFIDENTIAL

Technology Maturity: ENGINEERING_DEFINITION

Generated: 2026-08-28

Not physically validated. Not transfer-ready.

BUYER DECISION PAGE

WHAT IS IT?

Coat the external ventricular drain catheter with a local, sustained■release albendazole formulation to reduce proteinaceous debris and clot formation within the catheter lumen

WHY DOES IT MATTER?

External ventricular drain catheter occludes with blood clot and proteinaceous debris within days of insertion, causing dangerous intracranial pressure elevation

WHAT IS ACTUALLY ESTABLISHED?

Engineering definition with 3 governing equations, 12 design inputs, 6 failure modes. Computational model exists. No physical prototype.

WHAT IS NOT ESTABLISHED?

Physical validation, manufacturing process, regulatory pathway, IP ownership.

WHY MIGHT A BUYER CARE?

Opportunity in fluidics_hydraulic. 11-step build plan with specific test articles and acceptance criteria. Target buyer: domain-appropriate technology buyer (to be identified by portfolio owner).

WHAT DOES THE BUYER GET?

Buyer receives: canonical invention specification (with epistemic classes).; Buyer receives: engineering specification (this document, all blocks classed).; Buyer receives: design input/output graph with explicit IDs.; Buyer receives: engineering reasoning chains (claim->principle->model->input->assumption->output->failure->verification).; Buyer receives: killer-experiment definition and its EIG basis.; Buyer receives: evidence custody references and hashes.; Buyer receives: unresolved-uncertainty register.; Buyer receives: domain equation set (symbolic; sources and applicability judgments identified).

WHAT DOES THE BUYER HAVE TO BUILD?

Buyer must create: actual geometry and detailed design.; Buyer must create: sourced engineering parameters and acceptance thresholds.; Buyer must create: physical prototypes.; Buyer must create: all verification and validation results.; Buyer must create: manufacturing process qualification.; Buyer must create: regulatory pathway determination and submissions.

WHAT IS THE NEXT DECISIVE EXPERIMENT?

first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed)

WHAT WOULD MAKE US KILL IT?

killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.44573539382492045, EIG-per-cost 0.44573539382492045; the falsification test carries the pre-registered pass/fail before any transfer)

WHAT TRANSACTION COULD MAKE SENSE?

Sponsored Validation

What Exists Today

Existing:

- + Documented mechanism with governing equations
- + Computational model
- + 3 external evidence sources
- + Design inputs and outputs
- + Failure analysis
- + 11-step engineering build plan
- + Transfer boundary definition

Does Not Yet Exist:

- Physical prototype
- Qualified manufacturing process
- Clinical data
- Verified IP ownership
- Regulatory clearance
- Commercial product

Buyer Can Commission:

- > Specific bench experiment (see build plan)
- > Prototype fabrication
- > Materials study
- > Manufacturing feasibility study
- > Regulatory pre-submission

Development / De-risking Ladder

CURRENT STATE: ENGINEERING_DEFINITION

NEXT ENGINEERING STEP: first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed)

NEXT VALIDATION STEP: Bench prototype validation per build plan

NEXT REGULATORY STEP: Pre-submission to FDA for pathway classification

COMMERCIALIZATION GATE: Clinical validation evidence required

TRANSFER GATE: Transfer-ready when physical validation + regulatory pathway + IP verified

Engineering Investment Ladder

Next \$5K action: WP-01 - first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed) - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: physical bench/clinical experiment — NOT simulation. Uncertainty: Effort estimate, not quoted.

Next \$25K action: WP-03 - article for VF-001 - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: Conduct a randomized controlled trial comparing catheter patency rates and time to occlusion in patients receiving albendazole-coated cat.... Uncertainty: Effort estimate, not quoted.

Next \$100K action: WP-05 - article for VF-003 - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: bench flow-loop characterization across operating range. Uncertainty: Effort estimate, not quoted.

Top 5 Buyer Diligence Questions

Q1: Is the mechanism physically credible for the intended application?

The governing model includes 3 equations. Key assumptions: CSF approximated as Newtonian (valid at shunt flow rates); no wall compliance effects; transition criterion $Re \sim 2300$ is a MODEL_DERIVED engineering rule of thumb; $C_d \sim 0.6-0.65$ is an engineering prior, not a measurement; FLUID-001 (Hagen-Poiseuille (laminar pipe flow)) is applicable only under: Newtonian fluid, laminar regime, rigid circular lumen, fully developed flow.; FLUID-002 (Reynolds number) is applicable only under: circular section; single-phase flow.; FLUID-003 (Orifice flow (sharp-edged restriction)) is applicable only under: turbulent orifice flow; C_d treated as MODEL_DERIVED until measured.; Model input variables requiring sourced values before any numeric result: r (m), dP (Pa), μ (Pa.s), L (m), ρ (kg/m³), v (m/s), D (m), μ (Pa.s), C_d (-), A (m²), dP (Pa), ρ (kg/m³) — all UNKNOWN (no sourced value exists; Art. VIII).; Model output variables produced by the governing model: Q (m³/s), Re (-), Q (m³/s) — UNKNOWN until the inputs are sourced and the model is evaluated.; Critical parameters under specification (parameters with recorded units and epistemic status; no values invented): lumen inner diameter [r , m] value_status=UNKNOWN; lumen length [r , m] value_status=UNKNOWN; surface roughness [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; operating pressure head [dP , Pa] value_status=UNKNOWN; flow rate range [Q , m³/s] value_status=UNKNOWN; viscosity of flowed medium [μ , Pa.s] value_status=UNKNOWN; occlusion growth tolerance [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN.. UNKNOWN: whether assumptions hold in vivo.

Q2: What evidence supports the core claim?

3 external evidence sources. All are external precedent, not physical validation of this specific invention.

Q3: What is the regulatory pathway?

UNKNOWN. Not yet determined. Requires pre-submission to FDA. No predicate identified for some packages.

Q4: Who owns the IP?

UNKNOWN. IP ownership not verified. Requires legal counsel engagement.

Q5: What would it cost to validate?

First experiment: first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed). Estimated effort from build plan. Full validation cost: COST TO BE QUOTED.

Commercial Interest

TARGET APPLICATION: CSF shunt systems for hydrocephalus management

CURRENT ALTERNATIVE: Standard fixed-pressure or programmable CSF shunt valves

MECHANISM DIFFERENCE: Administration of albendazole reduces cystic fluid viscosity and protein content, potentially decreasing clot formation

POTENTIAL VALUE DRIVER: Addresses: External ventricular drain catheter occludes with blood clot and proteinaceous debris within days of insertion, causing dangerous intracranial pressure elevation

ADOPTION BARRIER: Requires physical validation, regulatory clearance, and manufacturing qualification

DEVELOPMENT BARRIER: No physical prototype exists; manufacturing process not established

Licensee Capability Fit

IDEAL BUYER: domain-appropriate technology buyer (to be identified by portfolio owner)

REQUIRED ENGINEERING CAPABILITY: Catheter design, fluid mechanics / domain-specific engineering

REQUIRED MANUFACTURING CAPABILITY: Medical tubing extrusion, injection molding, cleanroom assembly

REQUIRED REGULATORY CAPABILITY: FDA 510(k) or PMA submission experience

REQUIRED CAPITAL: Validation: \$50-200K. Full development: \$500K-2M (estimates, not quoted)

REQUIRED MARKET ACCESS: Neurosurgery market channels; hospital purchasing relationships

1. Technology Description

Domain: fluidics_hydraulic

Disciplines: fluid mechanics, mechanical engineering, biomedical engineering, materials science

Architecture: PROPOSED architecture (no physical system exists): flow path / lumen architecture; pressure regulation element; sensing / feedback element (if closed-loop); termination interfaces (patient / reservoir) — realizing: Coat the external ventricular drain catheter with a local, sustained■release albendazole formulation to reduce proteinaceous debris and clot formation within the catheter lumen

Subsystems:

?: flow path / lumen architecture - ?

?: pressure regulation element - ?

?: sensing / feedback element (if closed-loop) - ?

?: termination interfaces (patient / reservoir) - ?

2. Mechanism Architecture

Physical Changes: PROPOSED: Coat the external ventricular drain catheter with a local, sustained■release albendazole formulation to reduce proteinaceous debris and clot formation within the catheter lumen

Key Physics: Fluid mechanics / hydraulics governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

3. Governing Engineering Model

Summary: Fluid mechanics / hydraulics governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

Equations:

FLUID-001: Hagen-Poiseuille (laminar pipe flow). Expression: $Q = (\pi * r^4 * \Delta P) / (8 * \mu * L)$.

Applicability to this invention: APPLICABLE; holds under: Newtonian fluid, laminar regime, rigid circular lumen, fully developed flow. Assumptions: CSF approximated as Newtonian (valid at shunt flow rates); no wall compliance effects. Engaged variables: r , μ , L .

FLUID-002: Reynolds number. Expression: $Re = (\rho * v * D) / \mu$. Applicability to this invention: APPLICABLE; holds under: circular section; single-phase flow. Assumptions: transition criterion $Re \sim 2300$ is a MODEL_DERIVED engineering rule of thumb. Engaged variables: μ .

FLUID-003: Orifice flow (sharp-edged restriction). Expression: $Q = C_d * A * \sqrt{2 * \Delta P / \rho}$. Applicability to this invention: CONDITIONAL; holds under: turbulent orifice flow; C_d treated as MODEL_DERIVED until measured. Assumptions: $C_d \sim 0.6-0.65$ is an engineering prior, not a measurement. Engaged variables: NOT ESTABLISHED.

Assumptions:

- CSF approximated as Newtonian (valid at shunt flow rates)
- no wall compliance effects
- transition criterion $Re \sim 2300$ is a MODEL_DERIVED engineering rule of thumb
- $C_d \sim 0.6-0.65$ is an engineering prior, not a measurement
- FLUID-001 (Hagen-Poiseuille (laminar pipe flow)) is applicable only under: Newtonian fluid, laminar regime, rigid circular lumen, fully developed flow.
- FLUID-002 (Reynolds number) is applicable only under: circular section; single-phase flow.
- FLUID-003 (Orifice flow (sharp-edged restriction)) is applicable only under: turbulent orifice flow; C_d treated as MODEL_DERIVED until measured.

- Model input variables requiring sourced values before any numeric result: r (m), dP (Pa), μ (Pa.s), L (m), ρ (kg/m³), v (m/s), D (m), μ (Pa.s), C_d (-), A (m²), dP (Pa), ρ (kg/m³) — all UNKNOWN (no sourced value exists; Art. VIII).
- Model output variables produced by the governing model: Q (m³/s), Re (-), Q (m³/s) — UNKNOWN until the inputs are sourced and the model is evaluated.
- Critical parameters under specification (parameters with recorded units and epistemic status; no values invented): lumen inner diameter [r , m] value_status=UNKNOWN; lumen length [r , m] value_status=UNKNOWN; surface roughness [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; operating pressure head [dP , Pa] value_status=UNKNOWN; flow rate range [Q , m³/s] value_status=UNKNOWN; viscosity of flowed medium [μ , Pa.s] value_status=UNKNOWN; occlusion growth tolerance [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN.

Boundary Conditions:

- to be established by the first bench characterization (not yet performed)

Failure Regimes:

- {'regime': 'OCCLUSION', 'basis': 'SOURCE_FACT (problem.json)'}
- {'regime': 'proximal occlusion by tissue ingrowth', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'lumen collapse under external pressure (kink)', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'encrustation / mineral deposition', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'regime transition to turbulence altering calibration', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'check-valve seat wear opening reverse leakage', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}

4. Design Inputs

ID	Input	Value	Origin
DI-001	Stated constraint	Must maintain cerebrospinal fluid drainage patency for the required monitoring duration	SOURCE_FACT systemic anticoagulation
DI-002	Mechanism driver (proposed)	Administration of albendazole reduces cystic fluid viscosity and protein content	MODEL_EFFECT decreasing clot formation
DI-003	Expected effect target	Lower incidence of catheter occlusion by blood clot and proteinaceous debris	MODEL_EFFECT CSF drainage patency without
DI-004	Domain design input pattern: required flow applicability	Variable pressure [pressure, 'required']	ENGINEERING_PROPOSED
DI-005	Domain design input pattern: medium viscosity applicability	Fixed value [viscosity]	ENGINEERING_PROPOSED
DI-006	Domain design input pattern: anatomical applicability	NOT_ESTABLISHED (candidate pattern; confirm against patient data)	ENGINEERING_PROPOSED
DI-007	Domain design input pattern: device lifetime applicability	NOT_ESTABLISHED (candidate pattern; confirm against patient data)	ENGINEERING_PROPOSED
DI-008	Custodied observation: Continuous ventricular drainage	https://eur01.safelinks.com/m/4123d1822/content_hash/b1f8bd1f2ad8602803e4B423c24751cb5e2115e35b5f9b9	
DI-009	Custodied observation: Primary sensory cortex mapping	https://eur01.safelinks.com/m/4123d1822/content_hash/b1f8bd1f2ad8602803e4B423c24751cb5e2115e35b5f9b9	
DI-010	Custodied observation: Augmented reality / Surgical training	https://eur01.safelinks.com/m/4123d1822/content_hash/b1f8bd1f2ad8602803e4B423c24751cb5e2115e35b5f9b9	
DI-011	Custodied observation: Imaging findings of patient brain	https://eur01.safelinks.com/m/4123d1822/content_hash/b1f8bd1f2ad8602803e4B423c24751cb5e2115e35b5f9b9	
DI-012	Custodied observation: Neuroendoscopy / surgical exposure	https://eur01.safelinks.com/m/4123d1822/content_hash/b1f8bd1f2ad8602803e4B423c24751cb5e2115e35b5f9b9	

5. Design Outputs

ID	Description	Status	Missing Inputs
DO-001	flow path / lumen architecture realizing Coat the external ventricular drain catheter	CONFIRMED	detailed local sustained release albendazole formulation to reduce prote
DO-002	pressure regulation element realizing Coat the external ventricular drain catheter	CONFIRMED	detailed local sustained release albendazole formulation to reduce prote
DO-003	sensing / feedback element (if closed-loop) realizing Coat the external ventricular drain catheter	CONFIRMED	detailed catheter structure sustained release albendazole formulation
DO-004	termination interfaces (patient / reservoir) realizing Coat the external ventricular drain catheter	CONFIRMED	detailed catheter structure sustained release albendazole formulation to
DO-005	flow path / lumen architecture interfaces with pressure regulation element	CONFIRMED	interfacial material (off for Coats, external ventricular drain catheter
DO-006	pressure regulation element interfaces with sensing / feedback element (if closed-loop)	CONFIRMED	interfacial material (off for Coats, external ventricular drain catheter
DO-007	sensing / feedback element (if closed-loop) interfaces with termination interfaces	CONFIRMED	interfacial material (off for Coats, external ventricular drain catheter
DO-008	Device boundaries: patient/fluid inlet and outlet termination	CONFIRMED	control/data points required by Coats, external ventricular drain catheter with a l
DO-009	Flow/force/signal path geometry realizing the mechanism for the external ventricular drain catheter	UNKNOWN	tail chain/surface, as of, finished release albendazole formu
DO-010	Operating range for lumen inner diameter within which the mechanism holds its stated behavior	UNKNOWN	stated behavior computed bounds for lumen inner diameter; verification of ran
DO-011	Operating range for lumen length within which the mechanism holds its stated behavior	UNKNOWN	stated behavior computed bounds for lumen length; verification of range endp
DO-012	Operating range for surface roughness within which the mechanism holds its stated behavior	UNKNOWN	stated behavior computed bounds for surface roughness; verification of range
DO-013	Operating range for operating pressure head within which the mechanism holds its stated behavior	UNKNOWN	stated behavior computed bounds for operating pressure head; verification of
DO-014	Operating range for flow rate range within which the mechanism holds its stated behavior	UNKNOWN	stated behavior computed bounds for flow rate range; verification of range en
DO-015	Operating range for viscosity of flowed medium within which the mechanism holds its stated behavior	UNKNOWN	stated behavior computed bounds for viscosity of flowed medium; verification
DO-016	No closed-loop control is proposed: the invention operates as open-loop as specified; this record exists so the omission is explicit and auditable	CONFIRMED	
DO-017	Information path: sensing element -> decision/monitor	CONFIRMED	act (minimal); rich data paths are NOT APPLICABLE to this mechanical/physical in
DO-018	Fixture enabling: bench flow-loop characterization across operating range — rig concept	UNKNOWN	rig concept, build specification, acceptance instrumentation
DO-019	Fixture enabling: pressure-drop mapping vs flow (identifying transitions) — rig concept	UNKNOWN	rig concept, build specification, acceptance instrumentation
DO-020	Fixture enabling: accelerated occlusion / fouling challenge testing — rig concept	UNKNOWN	rig concept, build specification, acceptance instrumentation

ID	Description	Status	Missing Inputs
DO-021	Fixture enabling: pressure-flow mapping across full operating envelope — rig configuration, by: [redacted]	Planning	Conceptual design, instrumentation specification, acceptance instrumentation

6. Critical Design Parameters

Name	Value	Unit	Basis	Verification
lumen inner diameter	UNKNOWN (no sourced value)	m	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method
lumen length	UNKNOWN (no sourced value)	m	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method
surface roughness	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method
operating pressure head	UNKNOWN (no sourced value)	Pa	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method
flow rate range	UNKNOWN (no sourced value)	m^3/s	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method
viscosity of flowed medium	UNKNOWN (no sourced value)	Pa.s	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method
occlusion growth tolerance	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method

7. Failure Modes

Mode	Mechanism	Mitigation	Verification	Residual Uncertainty
OCCLUSION	Fails by OCCLUSION: the designation of a NOT-VALID STATE (requiring design change) is not being detected or acted in the problem (rectified by design change)	design change to NOT-VALID STATE (requiring design change) is not being detected or acted in the problem (rectified by design change)	design change to NOT-VALID STATE (requiring design change) is not being detected or acted in the problem (rectified by design change)	design change to NOT-VALID STATE (requiring design change) is not being detected or acted in the problem (rectified by design change)
proximal occlusion by tissue ingrowth	Proximal occlusion by tissue ingrowth	Proximal occlusion by tissue ingrowth	Proximal occlusion by tissue ingrowth	Proximal occlusion by tissue ingrowth
lumen collapse under external pressure	Lumen collapse under external pressure	Lumen collapse under external pressure	Lumen collapse under external pressure	Lumen collapse under external pressure
encrustation / mineral deposition	Encrustation / mineral deposition	Encrustation / mineral deposition	Encrustation / mineral deposition	Encrustation / mineral deposition
regime transition to turbulence	Regime transition to turbulence	Regime transition to turbulence	Regime transition to turbulence	Regime transition to turbulence
check-valve seat wear	Check-valve seat wear	Check-valve seat wear	Check-valve seat wear	Check-valve seat wear

8. Failure Analysis

Failure Mode	Mechanism	Mitigation	Residual Uncertainty
OCCLUSION	Fails by OCCLUSION: the device failure design controls NOT ESTABLISHED (e.g. when flow properties tested do not capture the problem) (Art. (X&Y))		
proximal occlusion by tissue ingrowth	Proximal occlusion by tissue ingrowth/geometric features surface dissolution (e.g. when flow properties tested do not capture the problem) (Art. (X&Y))		
lumen collapse under external pressure (collapse)	Lumen collapse under external pressure (less) external pressure (e.g. when flow properties tested do not capture the problem) (Art. (X&Y))		
encrustation / mineral deposition	Failure by encrustation / mineral deposition surface adhesion / coating degradation (e.g. when flow properties tested do not capture the problem) (Art. (X&Y))		
regime transition to turbulence	Failure by regime transition to turbulence altering geometry above the regime (e.g. when flow properties tested do not capture the problem) (Art. (X&Y))		
check-valve seat wear opening	Failure by check-valve seat wear opening seat leakage and repeated seating / closing (e.g. when flow properties tested do not capture the problem) (Art. (X&Y))		

9. Verification Strategy

ID	Requirement	Method	Acceptance	Result
VF-001	Conduct a randomized controlled trial comparing the device to control rates and to pre-registered endpoints (e.g. MACE, mortality, etc.) (Art. (X&Y))	Conduct a randomized controlled trial comparing the device to control rates and to pre-registered endpoints (e.g. MACE, mortality, etc.) (Art. (X&Y))	Acceptance criteria to be fixed at a later stage (e.g. MACE, mortality, etc.) (Art. (X&Y))	Not performed (definition; validation is required)
VF-002	run killer experiment: falsification_test_hunkilander (e.g. falsification_test_hunkilander) (Art. (X&Y))	run killer experiment: falsification_test_hunkilander (e.g. falsification_test_hunkilander) (Art. (X&Y))	Acceptance criteria to be fixed at a later stage (e.g. MACE, mortality, etc.) (Art. (X&Y))	Not performed (definition; validation is required)
VF-003	bench flow-loop characterization across operating range	bench flow-loop characterization across operating range	Acceptance criteria to be fixed at a later stage (e.g. MACE, mortality, etc.) (Art. (X&Y))	Not performed (definition; validation is required)
VF-004	pressure-drop mapping vs flow (identify pressure drops)	pressure-drop mapping vs flow (identify pressure drops)	Acceptance criteria to be fixed at a later stage (e.g. MACE, mortality, etc.) (Art. (X&Y))	Not performed (definition; validation is required)
VF-005	bench flow-loop characterization across operating range	bench flow-loop characterization across operating range	Acceptance criteria to be fixed at a later stage (e.g. MACE, mortality, etc.) (Art. (X&Y))	Not performed (definition; validation is required)
VF-006	pressure-flow mapping across full operating range	pressure-flow mapping across full operating range	Acceptance criteria to be fixed at a later stage (e.g. MACE, mortality, etc.) (Art. (X&Y))	Not performed (definition; validation is required)
VF-007	long-duration patency bench soak with long duration patency	long-duration patency bench soak with long duration patency	Acceptance criteria to be fixed at a later stage (e.g. MACE, mortality, etc.) (Art. (X&Y))	Not performed (definition; validation is required)
VF-008	pressure-flow mapping across full operating range	pressure-flow mapping across full operating range	Acceptance criteria to be fixed at a later stage (e.g. MACE, mortality, etc.) (Art. (X&Y))	Not performed (definition; validation is required)
VF-009	accelerated occlusion / fouling challenge testing	accelerated occlusion / fouling challenge testing	Acceptance criteria to be fixed at a later stage (e.g. MACE, mortality, etc.) (Art. (X&Y))	Not performed (definition; validation is required)

10. Validation Strategy

ID	Requirement	Method	Acceptance	Result
VA-001	end-use validation of verified design	NOT ESTABLISHED (defined after verification)	Acceptance criteria to be fixed at a later stage (e.g. MACE, mortality, etc.) (Art. (X&Y))	Not performed (definition; validation is required)

11. Materials

Component	Candidate Material	Source	Verification Required	Status
domain candidate material	medical-grade silicone elastomer	standard CSF shunt catheter material	True	ENGINEERING_PROPOSED
domain candidate material	polycarbonate / PSU rigid components	standard shunt valve housings	True	ENGINEERING_PROPOSED

12. Bill of Materials

Item	Description	Qty	Type	Material	Supplier	Criticality	Verification
1	medical-grade silicone elastomer	1	Material	Medical-grade silicone elastomer	Not Established	Not Established	Applicability verification required
2	polycarbonate / PSU rigid components	1	Material	polycarbonate / PSU rigid components	Not Established	Not Established	Applicability verification required

13. Manufacturing

Status: NO MANUFACTURING ANALYSIS POSSIBLE — no design exists; processes are candidates only

Candidate Processes:

- medical extrusion with inner-diameter tolerance control: domain registry manufacturing route - candidate only; no process qualification exists
- injection molding of valve/housing features: domain registry manufacturing route - candidate only; no process qualification exists
- laser drilling of calibrated orifices: domain registry manufacturing route - candidate only; no process qualification exists

14. External Evidence

Source 1:

URL :

Excerpt:

Source 2:

URL :

Excerpt:

Source 3:

URL :

Excerpt:

15. Transfer Boundary

YOU RECEIVE:

- + Buyer receives: canonical invention specification (with epistemic classes).
- + Buyer receives: engineering specification (this document, all blocks classed).
- + Buyer receives: design input/output graph with explicit IDs.
- + Buyer receives: engineering reasoning chains (claim->principle->model->input->assumption->output->failure->verification).
- + Buyer receives: killer-experiment definition and its EIG basis.
- + Buyer receives: evidence custody references and hashes.
- + Buyer receives: unresolved-uncertainty register.
- + Buyer receives: domain equation set (symbolic; sources and applicability judgments identified).

YOU MUST DEVELOP:

- > Buyer must create: actual geometry and detailed design.
- > Buyer must create: sourced engineering parameters and acceptance thresholds.
- > Buyer must create: physical prototypes.
- > Buyer must create: all verification and validation results.
- > Buyer must create: manufacturing process qualification.
- > Buyer must create: regulatory pathway determination and submissions.

YOU MUST VERIFY:

- ? Biocompatibility (ISO 10993)
- ? Sterilization compatibility
- ? Mechanical integrity
- ? Regulatory pathway
- ? IP ownership and freedom to operate
- ? Manufacturing process capability

16. Open Questions

- {'unknown': 'ALL sourced engineering parameter values', 'reason': 'no measurement or physical observation exists'}

17. Decisive Next Experiment

WP	Test Article	Equipment	Measurement	Acceptance	Deliverable	Effort
WP-01	first physical article	maintaining the proposed physical verification	conduct a randomized	NOT ESTABLISHED	first physical article	NOT ESTABLISHED
WP-02	constraint-compliance envelope article	per method	constraint-compliance	NOT ESTABLISHED	constraint-compliance envelope article	NOT ESTABLISHED (to be quoted)
WP-03	article for VF-001	per method	Conduct a randomized	NOT ESTABLISHED	VF-001 verification	NOT ESTABLISHED (to be quoted)
WP-04	article for VF-002	per method	run killer experiment:	NOT ESTABLISHED	VF-002 article (EIG INCOMPLETE)	NOT ESTABLISHED (to be quoted)
WP-05	article for VF-003	per method	bench flow-loop character	NOT ESTABLISHED	VF-003 article	NOT ESTABLISHED (to be quoted)
WP-06	article for VF-004	per method	pressure-drop mapping	NOT ESTABLISHED	VF-004 article	NOT ESTABLISHED (to be quoted)
WP-07	article for VF-005	per method	bench flow-loop character	NOT ESTABLISHED	VF-005 article	NOT ESTABLISHED (to be quoted)
WP-08	article for VF-006	per method	pressure-flow mapping	NOT ESTABLISHED	VF-006 article	NOT ESTABLISHED (to be quoted)
WP-09	article for VF-007	per method	long-duration patency	NOT ESTABLISHED	VF-007 article	NOT ESTABLISHED (to be quoted)
WP-10	article for VF-008	per method	pressure-flow mapping	NOT ESTABLISHED	VF-008 article	NOT ESTABLISHED (to be quoted)

WP	Test Article	Equipment	Measurement	Acceptance	Deliverable	Effort
WP-11	article for VF-009	per method	accelerated occlusion	NOT ESTABLISHED	NOT ESTABLISHED	NOT ESTABLISHED (to be quoted)

18. Buyer Decision Framework

WHY INVEST: Engineering definition exists with 3 governing equations and a 11-step build plan. The next experiment is clearly specified. Target buyer: domain-appropriate technology buyer (to be identified by portfolio owner).

WHY WAIT: No physical prototype exists. Regulatory pathway is unresolved. IP ownership is unverified. Manufacturing process is not qualified.

WHY REJECT: killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.44573539382492045, EIG-per-cost 0.44573539382492045; the falsification test carries the pre-registered pass/fail before any transfer)

WHAT WOULD CHANGE THE DECISION: A successful bench experiment demonstrating the core mechanism. A qualified regulatory pathway. A confirmed IP position.

19. Kill Condition

KILL_IF: killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.44573539382492045, EIG-per-cost 0.44573539382492045; the falsification test carries the pre-registered pass/fail before any transfer)

DISCLOSURE

This dossier distinguishes established evidence, external precedent, computational/modelled results, engineering proposals, and unresolved questions. The absence of physical validation should not be interpreted as evidence that the technology will fail; it means the relevant proposition has not yet been established experimentally.

This dossier does not constitute legal, patentability, FTO, regulatory, or investment advice.